Physical chemistry

→ Electrical properties of molecules

⇒

Electrical properties of molecules

2018

(a) Define the terms (i) induced polarization (ii) orientation polarization (iii) polarisability. How can you distinguish between polar and non polar molecules from polarization measurement

(b) Derive an expression for the polarization of dielectrics in the gas phase and hence the mosotti-clausius equation for the non-polar molecules

(c) Name the major applications of the dipole moment measurements. Describe any one of them

(d) Two HCl molecules ($\mu = 1.08$ D) are separated by 4.0 Å (400pm) in air. Calculated the dipole-dipole interaction energy in kJmol^{-1}. If they are oriented end-to-end, that is, H-Cl-H-Cl. Given ($\varepsilon_0 = 8854 \times 10^{-12}$ C^{2}N^{-1}m^{-2})

2017:

(a) Same as 2018 (a)

(b) Derive an expression for the Debye equation

(c) Same as 2018(c)

(d) In water molecules two identical O-H bonds each of bond moment of 1.5 debye. Calculated the bond angle.

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2016:

(a) Define the term polarizability

(b) Explain the term distortion polarization and orientation polarization

(c) Derive an expression for clausius - mosalti equation

(d) Discuss the application of dipole moment.

(e) The dissociation energy of a molecule is 192.95 kcal.mol^{-1} what is the largest wavelength of light capable of dissociation of the molecule

2020:

(a) what do you mean by dipole moment? calculate the dipole moment if an electron is placed 1Å^0 from an equal positive charge

(b) why So_2 is polar and Co_2 is non polar

(c) Derive an expression for Debye equation

(d) Discuss about the application of dipole moment

(e) The dipole moment and polarizability of chlorobenzen are 1.57 D and 1.23×10^{-23} cm^{-3} respectively and the density at 25°C is 1.1171 gcm^{-3}. calculated the dielectric constant of the chlorobenzene

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2021:

(a) what do you mean by polarizibility

(b) Explain the terms distortion polarization and orientation polarization

(c) Derive an expression for clasius- mosotti equation

(d) Describe the Ebert's method and vapour- temperature method for the determination of dipole moment

(e) The total molar polarization of CH_3Cl is 90.12 cc/mole at $1/T = 0.004$ and 50.0 cc/mole extrapolated to $1/T = 0.0$. calculated the polarizability and dipole moment for CH_3Cl.

B.Sc. (Hons.) Part-III Tutorial Examination 2022
Course: Chem. 310, Marks: 20, Time: 1 hrs.

32.16 / 85

1.	What do you mean by polar and nonpolar molecules? Differentiate between polar and nonpolar molecules in terms of molar polarization.	3.0
2.	Define the terms polarization and polarizability.	1.5
3.	Derive an expression for Debye equation.	5.0
4.	Explain dipole-dipole, dipole-induced dipole and induced dipole-induced dipole interactions.	4.0
5.	Discuss the applications of dipole moment.	3.5
6.	At 0^0C, the molar polarization of a liquid (M = 85 g mol^{-1}) is 32.16 cc mol^{-1} and its density is 1.92 g cc^{-1}. Calculate the (a) dielectric constant value, and (b) polarizability of that compound.	3.0

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CT-2021

(i) Define the terms: polarization and polarizibility 2

(ii) Explain the dependence of polarizability on frequency. 4

(iii) Derive an expression for clausius-mosotti equation 4

(iv) Describe the Ebert's method and vapour-temperature method for the determination of dipole moment. 4

(v) The total molar polarization of CH_3Cl is 90.12 cc/mole at $1/T = 0.004$ and 50.0 cc/mol extrapolated to $1/T = 0.0$. Calculated the polarizability and dipole moment for CH_3Cl 3

(vi) for SO_2 gl the dielectric constant and density at room temperature are 1.00989 and 2.6287 kg/m^3 respectively. This gas has a permanent dipole moment of 1.62 D. Calculated (a) orientation polarization (b) total polarization and (c) total polarizability of SO_2.

CT-2019

(i) Consider a diatomic molecule AB. The observed and calculated values of dipole moment of this molecule are respectively 6.32 D and 7.93 D if $x_A = 4.0$ Calculated x_B.

(ii) The dipole moment of NH_3 molecules is 1.46 D. If the angle $\angle HNH = 108°$. Calculated the dipole moment of NH bond.

(iii) A Sodium ion (rat) is situated in air at a distance of 4.0 A° (400 pm) from a HCl molecule with a dipole moment of 1.08 D. Calculated the potential energy of interaction in kJ/mol ($q = 1.602 \times 10^{-19}$) $\epsilon = 8.854 \times 10^{-12}$

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Polar molecule: Polar molecule is a chemical species in which the distribution of electrons between the covalently bonded atoms is not even.

→ Polar molecules occur when there is an electronegativity difference between the bonded atoms.

→ So the molecules with a **non-zero net dipole moment** are called a polar molecules.

→ HCl, CH_3Cl, NH_3, H_2O, SO_2, C_2H_5OH etc are polar molecules.

Non-Polar molecules:

Non-polar molecules is a chemical species in which the distribution of electrons between the covalently bonded atoms is even.

→ Non-polar molecules occur when electrons are shared equal between atoms of a diatomic molecule

→ So the molecules with **a zero net dipole moment** are called a non-polar molecules

→ H_2, Cl_2, CH_4, C_6HC etc are non-polar molecules

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Bond moment or bond dipole moment:

✗ The degree of polarity of a chemical bond as calculated from the value of the force of the response of the bond when the bond is subjected to an electric field is called as bond dipole moment.

→ This occurs whenever there is a separation of positive and negative charges due to the unequal attraction of the bonded atoms.

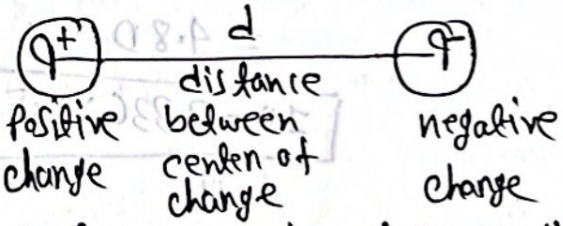
→ Any bond which has a degree of polarity has a dipole moment. This is called bond moment.



Dipole moment: Dipole moment is the product of the magnitude of the charges and the distance between them.

→ if 'q' is the magnitude of the charge and 'd' is the distance between the charges. Then the dipole moment μ is given by

$$\vec{\mu} = q \cdot \vec{d}$$



→ Dipole moment occurs due to the difference in electronegativities of two atoms in a bond. Greater the difference in electronegativity greater the dipole moment and greater the degree of polarity of bond.

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→ if there are two or more bonds in a molecule, the net dipole moment will be the vector addition of dipole moments of chemical bonds.

$$\vec{\mu} = \sum q_i \vec{d_i}$$
$$= \mu_1 + \mu_2 + \mu_3 + \cdots$$

→ if an electron is placed $1A^0$ from an equal positive charge.

$$\mu = q \times d$$
$$= 4.8 \times 10^{-10} \text{ esu} \times 10^{-8} \text{ cm}$$
$$= 4.8 \times 10^{-18} \text{ esu-cm}$$

q = charge of electron 4.8×10^{-10} esu

d = distance $1A^0$ $= 10^{-8}$ cm

The quantity 10^{-18} esu-cm is called one debye (D)

$$\boxed{1D = 10^{-18} \text{ esu-cm}}$$

So $\mu = 4.8 D$

in SI System:

$$\mu = q \times d$$
$$= 1.6 \times 10^{-19} C \times 10^{-10} m$$
$$= 1.602 \times 10^{-29} \text{ cm}$$
$$\simeq 4.8 D$$

$$\boxed{1D = 3.336 \times 10^{-30} \text{ cm}}$$

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Bond moment

→ Bond moment occurs in covalent chemical bond

→ Bond moment is the individual dipole moment in a molecule

→ In CCl_4 the bond moment between C-Cl is 1.5D

dipole moment

→ Dipole moment occurs between two ions in an ionic bond or between two atoms in a covalent bond

→ individual bond moments in a molecule add up to give overall dipole moment of the molecule

→ In CCl_4 the ormall dipole moment is Zero

→ The polarity of the molecule is measured by dipole moment

Polar bond: A Polar bond is defined by, where the bond forming electrons are unequally distributed between two atoms when one end is slightly positive and one end is slightly negative

$$H^{\delta+} \!\!-\!\!- Cl^{\delta-}$$

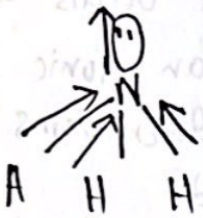
non-Polar bond: A non-Polar bond is defined by where the bond forming electrons are equally distributed between two atoms.



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⇒ Dipole moment of NF_3 is much less than NH_3



Result dipole moment = 1.46D



Resultent dipole moment = 0.24D

In NH_3, N is more electronegative than H. So 'N' pulls the electron from H towards itself and so the direction of bond moment and lone pair of e^- are in same direction, Hence dipole moment increased

on the other hand, In NF_3, N is less electronegative than F. So F atoms pulls the electron from N towards themselves. And the direction of bond moment and lone pair of electrone are diffrent so the resultent dipole moment is decreased

⊛ why SO_2 is polar but CO_2 and CCl_4 is non-polar?

SO_2 is Polar molecule Polar molecule is a chemical species in which the distribution of electrons between the covalent bonded atom is not even.

The electronegativity of O is 3.5 and Sulfur is 2.5 between them electronegativity is 1. we know if electronegativity

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difference (0.5-1.9) this molecule is polar because one end is slightly positive and other end is slightly negative.

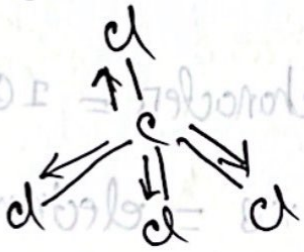


In SO_2, O is the more electronegative than S, so 'O' pulls the electron from S towards itself and so the direction of bond moment and lone pair of e^- are in same direction. So dipole moment increased and SO_2 is more polar

CCl_4 is non-polar molecule:

Non-polar molecules is a chemical species in which the distribution of electrons between the covalently bonded atoms is even.

In CCl_4, The four bond in CCl_4 are equal in the magnitude, Every Cl Bond is polar because the Cl atom is more electronegative than C atom, But their point in opposite direction and they are cancle each other. The sum of resultant dipole moment could be zero. Hence CCl_4 is non polar molecule



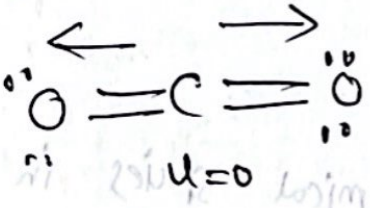
$\mu = 0$

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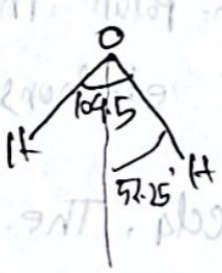
CO_2 is non polar molecule: Non-polar molecule is a chemical species in which the distribution of electron between the covalently bonded atoms is even.

The two bond in CO_2 are equal in the magnitude, Every C=O bond is polar because the 'O' atom is more electronegative than 'C' atom. But their point in opposite direction and they are cancle each other. The sum of resultant dipole moment could be zero, so CO_2 is non polar molecule



Problem

(*) measured dipole moment of water = 1.85 D



we know,

$\mu_{obs} = 2\mu_{OH} \cos\theta$

$\Rightarrow 1.85 = 2\mu_{OH} \cos 52.25°$

$\Rightarrow 1.85 = 2\mu_{OH}\ 0.6129$

$\Rightarrow \mu_{OH} = 1.51D$

$\Rightarrow$ percent ionic character $= 16(\chi_A - \chi_B) + 3.5(\chi_A - \chi_B)^2$

χ_A and χ_B = electronegativity of atom A and B

$\Rightarrow$ Electronegativity is the ability of an atom to attract toward itself the electrons in a chemical bond.

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When polar molecules are placed in an electric field. two effect will arise

① Induced or Distortion polarization:

The creation of positive and negative charges in a neutral molecule under the influence of electric field is called electrical or distortion or **induced polarization.**

→ Induced polarization is temporary and disappears as the electric field is removed.

→ This type of dipole moment is called induced dipole moment and dipole formed is called induced dipole.

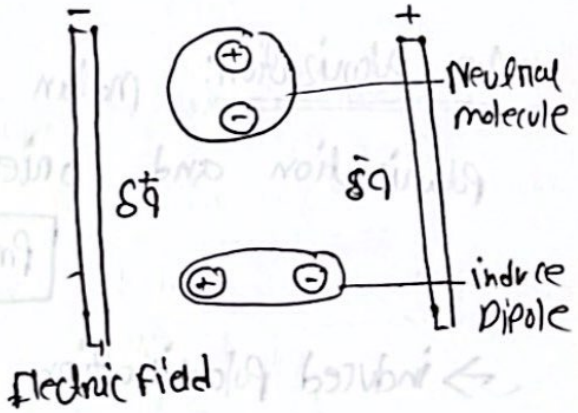


Fig: induced polarization of molecules

There are two types of induced polarization

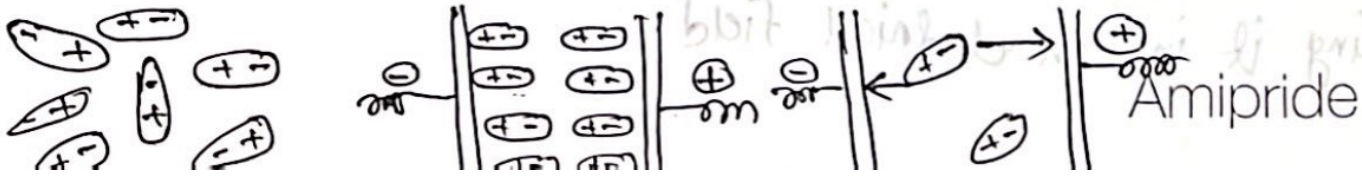
Electric polarization (Pe): electron will be distorted with respect to the nuclei towards the positive plates

Atomic polarization (Pa): The nuclei will be distorted with respect to each other.

So The induced polarization $P_i = P_e + P_a$

⑪ Orientation polarization:

Due to the thermal agitation, molecules occupy mean position between the direction of electric field and original position of molecules This is called orientation polarization.



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Polar molecules are randomly oriented in all directions, in absence of electric field.

if the polar molecules are stationary, the would orient at 180° to the direction of electric field.

oriented

Polarization

Molar Polarization: Molar polarization is the sum of the induced polarization and orientation polarization. it is denoted by P_m

$$P_m = P_i + P_o$$

- Induced polarization is temperature independ depends on the ratio of dielectric constant and density
- Orientation polarization is temperature dependent $P_o \propto \frac{1}{T}$

Molar polarization of polar molecule

(i) The molar polarization of polar molecules have permanent dipole moment

(ii) They have orientation polarization

(iii) They are temperature dependant

(iv) Example: HCl, CH_3Cl, $C_6H_5NO_2$

Molar Polarization of non polar molecule

(i) The molar polarization of non-polar molecules do not have permanent dipole moment

(ii) They have no orientation polarization

(iii) They are temperature independant

(iv) Example: O_2, CO_2, N_2

(7) **Polarizability:** Polarizability is the ability to form instantaneous dipoles. we obtain structural information about a molecule by placing it in an electrical field:

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Electric field has induced dipole moment so the induced dipole moment (μ_i) is directly proportional to the intensity of electric field (F_i)

Here $\mu_i \propto F_i$

$\Rightarrow \mu_i = \alpha F_i$

where, α is called polarizability

Polarization: Polarization is the displacement of positive charges relative to negative charges in a system.

$P = N\mu_i$ $\quad \mu_i = \alpha E_i$

$P = N\alpha F_i$ $\quad N = \text{molecules}/cm^3$

Clausius-Mosotti Equation:

The relation bdween dielectric constant and induced polarization is called Clausius-Mosotti equation.

when polar or non-polar molecule is placed in an electric field there will be small displacement of the electrical center which will be induced in the molecules

if μ_i is the induced dipole moment and F is the intensity of electric field. So

$$\mu_i = \alpha_0 F \quad -(1)$$

Suppose the strength of a uniform electric field produced by the two charges plates is F_0. then in any non-polar medium the field strength is reduced to E. So we can write.

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$$E_0 = E + 4\pi I \quad \text{--} (11)$$

$$\Rightarrow E_0 - E = 4\pi I$$

$$\Rightarrow E\left(\frac{E_0}{E} - 1\right) = 4\pi I$$

$$\Rightarrow E(\epsilon - 1) = 4\pi I \quad \text{---} (111)$$

where ϵ is called the dielectric constant of the medium and I is the induced electric moment per unit volume.

$$I = \mu_i n \quad \text{--} (IV)$$

In case of liquids and gases, the molecules are oriented in the absence of an external field.

$$F = E_0 + \frac{4}{3}\pi I - 4\pi I \quad \text{--} (V)$$

introducing equation (2) in (5) we get,

$$F = E + 4\pi I + \frac{4}{3}\pi I - 4\pi I$$

$$\therefore F = \frac{4}{3}\pi I + E \quad \text{--} (VI)$$

introducing equation (3) in (6) we get,

$$F = E + \frac{4}{3}\pi \frac{E(\epsilon - 1)}{4\pi}$$

$$= E + \frac{1}{3} E(\epsilon - 1)$$

$$= E + \frac{1}{3} E\epsilon - \frac{1}{3} E$$

$$= E\left(1 - \frac{1}{3}\right) + \frac{1}{3} E\epsilon$$

$$= \frac{2E}{3} + \frac{1}{3} E\epsilon$$

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$$= \frac{1}{3}(2E + E\epsilon)$$

$$\therefore F = \frac{E}{3}(2+\epsilon) \quad \text{--- (VII)}$$

Substituting equation (7) in (1)

$$\Rightarrow \frac{\mu_i}{\alpha_D} = \frac{E}{3}(2+\epsilon)$$

$$\Rightarrow \mu_i = \alpha_D \left(\frac{E}{3}\right)(2+\epsilon) \quad \text{--- (VIII)}$$

Substituting equation (VIII) in (4) we get

$$\frac{I}{n} = \alpha_D \left(\frac{E}{3}\right)(2+\epsilon)$$

$$\Rightarrow I = \frac{E}{3}(2+\epsilon) n\alpha_D \quad \text{— (IX)}$$

Substituting equation (9) in (3), we get

$$\frac{F(\epsilon-1)}{4\pi} = \frac{E}{3}(2+\epsilon) n\alpha_D$$

$$\Rightarrow \frac{\epsilon-1}{\epsilon+2} = \frac{4\pi}{3} n \cdot \alpha_D \quad \text{— (X)}$$

if ρ is the density of the medium between the charged plates and m is the molecular weight. then the number of molecules n in unit volume is $\frac{N\rho}{m}$, where N is the avogadros number's

$$n = \frac{N\rho}{m} \quad \text{--- (XI)}$$

Substituting equation (11) in (10) we get

$$\frac{\epsilon-1}{\epsilon+2} = \frac{4\pi}{3} \frac{N\rho}{m} \alpha_D$$

$$\Rightarrow \boxed{\frac{\epsilon-1}{\epsilon+2} \cdot \frac{m}{\rho} = \frac{4\pi}{3} N\alpha_D}$$

~~This is clausius mossotti equation~~

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The left hand side of the equation is given by the symbol P_m is called molar polarization of the material.

$$P_m = \frac{\epsilon - 1}{\epsilon + 2} \frac{M}{\rho} = \frac{4\pi}{3} N\alpha_D$$

This is clausius- mosotti equation

$$P_i = P_m = \frac{\epsilon - 1}{\epsilon + 2} \frac{M}{\rho} = \frac{4\pi}{3} N\alpha_D$$

Debye Equation:

The relation between molar polarization and temperature is called Debye equation.

when polar molecules are placed in an electric field the molecules orient themselves in the direction of the field.

The electric moment (μ_m) per molecule is given by

$$\mu_m = \alpha F$$

$$\text{or, } \mu_m = (\alpha_D + \alpha_0) F$$

$$\Rightarrow \mu_m = \alpha_D F + \alpha_0 F \quad \text{—(i)}$$

$$\Rightarrow \mu_m = \mu_i + \bar{m} \quad \text{—(ii)}$$

where, μ_i is the induced polarization and $\bar{m}$ is the orientation polarization

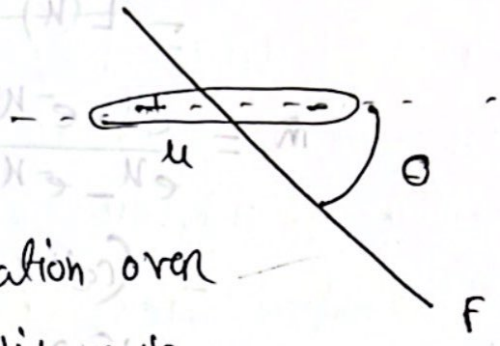
if μ is the permanent molecular moment and molecular dipoles are oriented that their axis make an angle θ with the direction of the field. the potential energy of a molecule is

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given by $U = -\mu F\cos\theta$ --(19)

Now according to Boltzmann's law, the number of molecules with their dipoles axes in the direction with a solid angle (dw) is given by

$$dN = Ae^{-\frac{U}{KT}} dw$$



The total number of molecules can be obtained by integrating by the above equation over the possible direction. for the solid angle between 0 and 2π

$$\text{Total no of molecules} = \int_0^{2\pi} Ae^{-\frac{U}{KT}} dw$$

$$N = \int_0^{2\pi} Ae^{\frac{\mu F\cos\theta}{KT}} dw$$

$$\text{Total moment} = \int_0^{2\pi} Ae^{\frac{\mu F\cos\theta}{KT}} \mu\cos\theta\, dw$$

The average moment per molecule $\bar{m}$ in the direction of the field is

$$\bar{m} = \frac{\int_0^{2\pi} Ae^{\frac{\mu F\cos\theta}{KT}} \cdot \mu\cos\theta\, dw}{\int_0^{2\pi} Ae^{\frac{\mu F\cos\theta}{KT}} \cdot dw}$$

Substituting $\frac{\mu F}{KT} = x$ and $\cos\theta = t$ and making use of the relation $dw = 2\pi\sin\theta\, d\theta = 2\pi dt = dt$

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$$\bar{m} = \frac{\mu \int_0^{2\pi} \ell e^{\ell\mu} d\ell}{\int_0^{2\pi} e^{\ell\mu} d\ell}$$

$$= L(\mu)\,\mu \quad [\text{Langevin function}]$$

$$\bar{m} = \frac{e^{\mu}+e^{-\mu}}{e^{\mu}-e^{-\mu}} - \frac{1}{\mu}$$

$$= \left(\coth\mu - \frac{1}{\mu}\right)$$

$$= L(\mu) \quad \text{---(IV)}$$

where $L(\mu)$ is the **Langevin function**, if x is small particularly for small field strength it is found that $L(\mu) \approx \frac{1}{3}\mu$

$$m = \frac{1}{3}\mu x = \frac{\mu^2 f}{3kT} = \frac{1}{3}\mu\mu \quad \text{---(V)}$$

on comparing equation (I) and (II)

$$\bar{m} = \alpha_0 f \quad \text{---(VI)}$$

Substituting equation (V) and (VI)

$$\alpha_0 f = \frac{\mu^2}{3kT} f$$

$$\alpha_0 = \frac{\mu^2}{3kT}$$

But the total polarisation, α is given by

$$\alpha = \alpha_D + \alpha_0$$

$$= \alpha_D + \frac{\mu^2}{3kT} \quad \text{---(VII)}$$

The molar

$\frac{\epsilon-1}{\epsilon+2} \cdot \frac{m}{\rho}$

orientation

if α

Sub

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The molar polarisation, which is assumed from dielectric constant $\frac{\epsilon-1}{\epsilon+2} \cdot \frac{M}{\rho}$ is the sum of the two effects. induced and orientation polarisation

$$P_m = P_i + P_o$$

$$P_m = \frac{\epsilon-1}{\epsilon+2} \cdot \frac{M}{\rho}$$

if α represents the total polarisability. ~~the equation~~

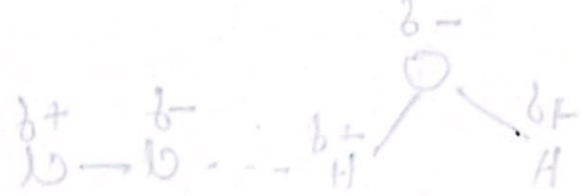
$$\frac{\epsilon-1}{\epsilon+2} \cdot \frac{M}{\rho} = \frac{4\pi N}{3} \alpha \quad \text{— (VIII)}$$

Substituting equation (VII) in (VIII)

$$\frac{\epsilon-1}{\epsilon+2} \frac{M}{\rho} = \frac{4\pi N}{3} \left(\alpha_D + \frac{\mu^2}{3KT}\right) \qquad [P_m = R_t]$$

$$\Rightarrow P_m = \frac{4\pi N}{3} \alpha_D + \frac{4\pi N}{9} \frac{\mu^2}{KT}$$

This is called Debye equation.



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Dipole-Dipole interactions:

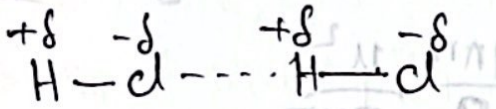
→ Dipole-dipole interactions are electrostatic interactions between molecules which have permanent dipoles

→ Dipole-dipole interactions are attractive forces between the positive end of one polar molecule and the negative end of another polar molecule.

→ Dipole-dipole interactions occur between molecules with permanent dipoles.

→ The hydrogen bond is a special type dipole-dipole interactions. This interactions is stronger than the London forces but is weaken than ion-ion interaction. HF > HI > HBr > HCl

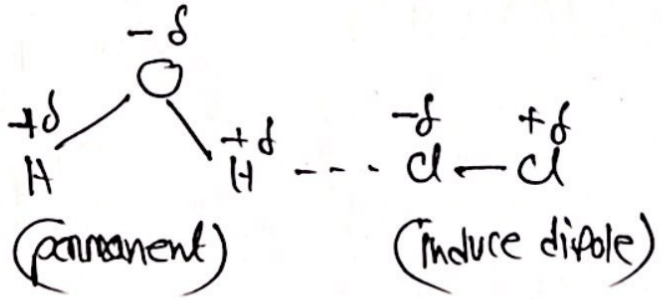
→ Dipole-dipole interactions are also known as Keesom interactions



Dipole-induced dipole interactions:

→ Dipole-induced dipole interactions occurs between a polar molecule and a non-polar molecule or atom. polar molecules causes temporary polarization of nearby non-polar molecule. this interaction is called Debye force.

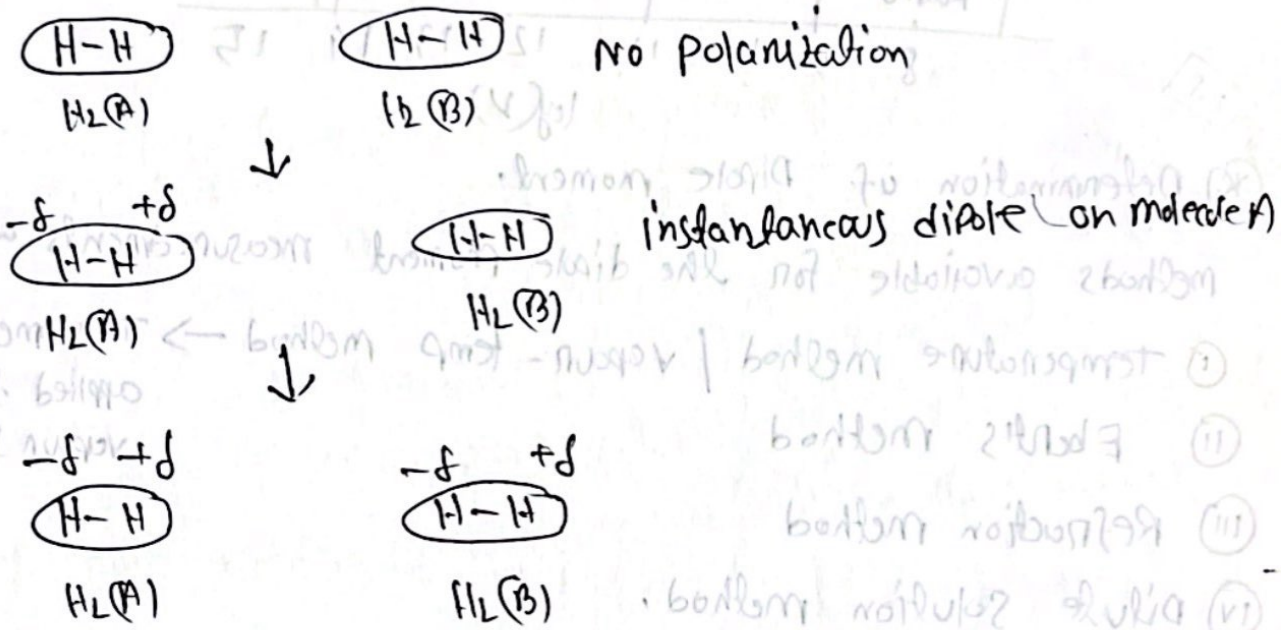
Example:



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induced dipole-induced dipole interaction:

induced dipole-induced dipole interaction is actually known as **London force**. The London force is a temporary attractive force that results when **the electrons in two adjacent atoms occupy positions** that make the atoms form temporary dipoles.



✱ Dependence of polarization on frequency:→

① Debye equation is derived assuming molecules in static electric field

② when applied field is oscillating, Debye equation can not be applied when the frequency is very high.

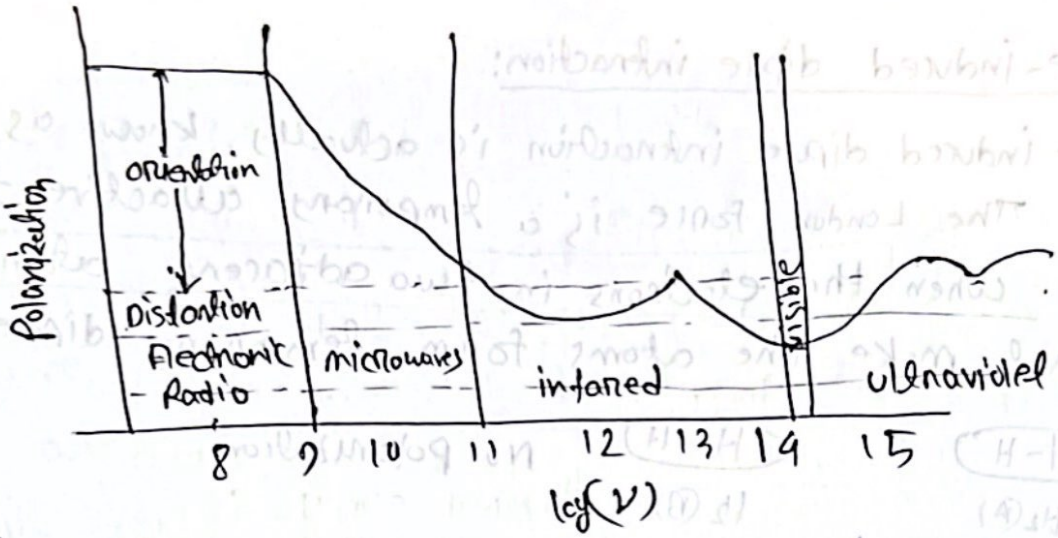
③ permanent dipole moment can not align with the ever changing electric field.

④ At high frequencies permanent dipole moment do not make any contribution to polarization.

⑤ A molecule needs nearly a pico-second (10^{-12} s) to rotate in a Liquid.

Hexinor

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(*) Determination of Dipole moment.

methods available for the dipole moment measurements are

(i) temperature method / vapour - temp method → This method is applied for gaseous / vapour state.

(ii) Ebert's method

(iii) Refraction method

(iv) Dilute solution method.

Temperature method: By determining the dielectric constant directly. The total polarisation (P_t) of a substance is obtained. Made being determined separately. We can write

$$P_t = a + \frac{b}{T}$$

if total polarisation P_1 and P_2 are determined at two temperature T_1 and T_2 respectively then

$$P_1 = a + \frac{b}{T_1} \quad \text{and} \quad P_2 = a + \frac{b}{T_2}$$

now,

$$P_1 - P_2 = a + \frac{b}{T_1} - a - \frac{b}{T_2}$$

$$P_1 - P_2 = \frac{b}{T_1} - \frac{b}{T_2}$$

$$\Rightarrow P_1 - P_2 = b\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

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$$\therefore b = (P_1 - P_2)\left(\frac{T_1 T_2}{T_2 - T_1}\right) \quad \text{--- (1)}$$

The value of dipole moment μ is easily calculated from that of b

$$b = \frac{4\pi N \mu^2}{9K}$$

$$\Rightarrow \mu = \sqrt{\left(\frac{9K}{4\pi N}\right) \cdot b}$$

As $K = 1.3807 \times 10^{-16}$ erg. K^{-1}, $\pi = 3.1416$, $N = 6.023 \times 10^{23}$ we get

$$\mu = \sqrt{\frac{9 \times 1.3807 \times 10^{-16} \cdot b}{4 \times 3.14 \times 6.023 \times 10^{23}}}$$

$$\mu = 0.0128 \times 10^{-18} \sqrt{b} \text{ esu. cm}$$

$$\mu = 0.0128 \times \sqrt{b}\ D$$

So knowing b from equation (1) we can calculated the dipole moment easily.

Ebert's method's: Polar molecules in the gaseous state when subjected to an electric field will undergo both orientation and distortion polarisations. But when the same substance in the solid state is subjected to the same field, the orientation of molecules is not possible and only distortion polarisation will occur. Therefore the total molar polarisation is given by

$$P_{gas} = P_i + P_o$$

$$P_{solid} = P_i$$

orientation polarisation $P_o = P_{gas} - P_{solid}$

Amipride

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$$\frac{4}{3}\pi N\left(\frac{\mu^2}{3KT}\right) = P_{gas} - P_{solid}$$

$$\mu = \sqrt{\frac{9KT}{4\pi N}\left(P_{gas} - P_{solid}\right)} \quad --(1)$$

As all the values of equation (1) are known we can calculated the value of dipole moment.

(*) Application of Dipole moment →

(i) The dipole moment helps to predict whether a molecule is polar or non-polar

(ii) The percentage of ionic character can be calculated from dipole moment

(iii) it helps to distinguish between cis- and trans- isomers

(iv) it helps to distinguish between ortho, meta and para- isomers

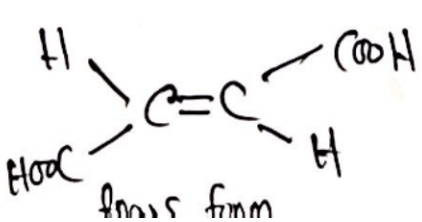
(v) Bond angle can be calculated with the help of it

(vi) it helps to determine the resonance structure

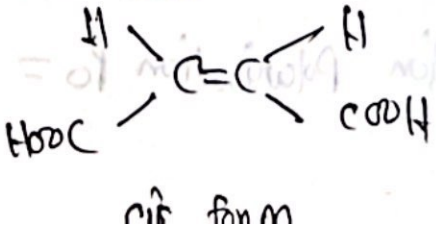
(vii) it helps to determine the shape of molecules.

Distinction between cis- and trans- isomer:

→ Dipole moments help us in distinguishing between the cis- and trans- isomers of a compound.



trans form



cis form

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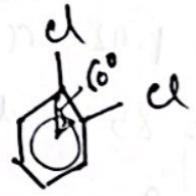
Information regarding aromatic substitution:

→ Benzene has a dipole moment zero

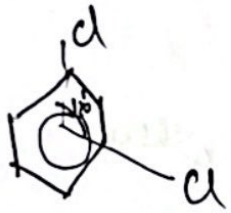
→ Thus it is a planner regular hexagon

→ These calculated values of dipole moment agree well with the experimental values

→



$\mu = 2.0D$



$\mu = 1.5D$



$\mu = 0$

The above structures of o-, m- and p- isomers are correct.

Determination of shape of molecules:

→ Inorganic Compounds

→ Monoatomic gases → He, Ne

→ Diatomic molecules.

- Homonuclear N_2, Cl_2
- Heteronuclear HCl, CO

→ Static molecules : O=C=O, S=C=S

Hexinar

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(*) At 0°C, the molar polarization of a liquid ($M = 85\ g\ mol^{-1}$) is 32.16 $cc\ mol^{-1}$ and its density is $1.92\ g\ cc^{-1}$. Calculate the

(i) dielectric constant value (ii) polarizabilities of the compound.

$$\frac{\epsilon-1}{\epsilon+2}\frac{M}{\rho} = \frac{4\pi N}{3}\left(\alpha_D + \frac{\mu^2 f}{3kT}\right)$$

$$\frac{\epsilon-1}{\epsilon+2}\frac{M}{\rho} = \frac{4\pi N}{3}\alpha_D$$

$P_m = 32.16\ cc\ mol^{-1}$

$\rho = 1.92\ g\ cc^{-1}$

$M = 85\ g\ mol^{-1}$

$\epsilon = ?$

$\alpha_0 = ?$

$\alpha = \alpha_0 + \alpha_0$ $\quad P_m = P_i + P_o =$ $\quad T = \infty$ $\quad P_o = 0$

$$P_m = \frac{\epsilon-1}{\epsilon+2}\cdot\frac{M}{\rho} = \frac{4\pi N}{3}\left(\alpha_0 + \frac{\mu^2}{3kT}\right)$$

$P_m = P_i$

$\Rightarrow 32.16 = \frac{\epsilon-1}{\epsilon+2}\cdot\frac{85}{1.92}$

$\Rightarrow \frac{\epsilon-1}{\epsilon+2} = \frac{32.16 \times 1.92}{85}$

$\Rightarrow \epsilon = 9.11$

$\Rightarrow \frac{\epsilon-1}{\epsilon+2}\cdot\frac{M}{\rho} = \frac{4\pi N}{3}(\alpha_D + \alpha_0)$

6.02×10^{23}

$\Rightarrow 32.16 = \frac{4\pi N \alpha_0}{3}$

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